



D.I.R.T.T. (Digital Instrument for Remote Terrain Testing)

Nolan Bretchel^{PM,1}, George Utz^{CS,2}, Cara Belew¹, Zac Morrow¹, Rebecca Ueltschey²

Dr. Christopher Hudson^{A,3}

^{PM} – Project Manager

^{CS} – Chief Scientist

^A – Faculty Advisor

^{EA} – External Advisor

¹ – Computer Engineering

² – Electrical Engineering

³ – Computer Science

The Problem:

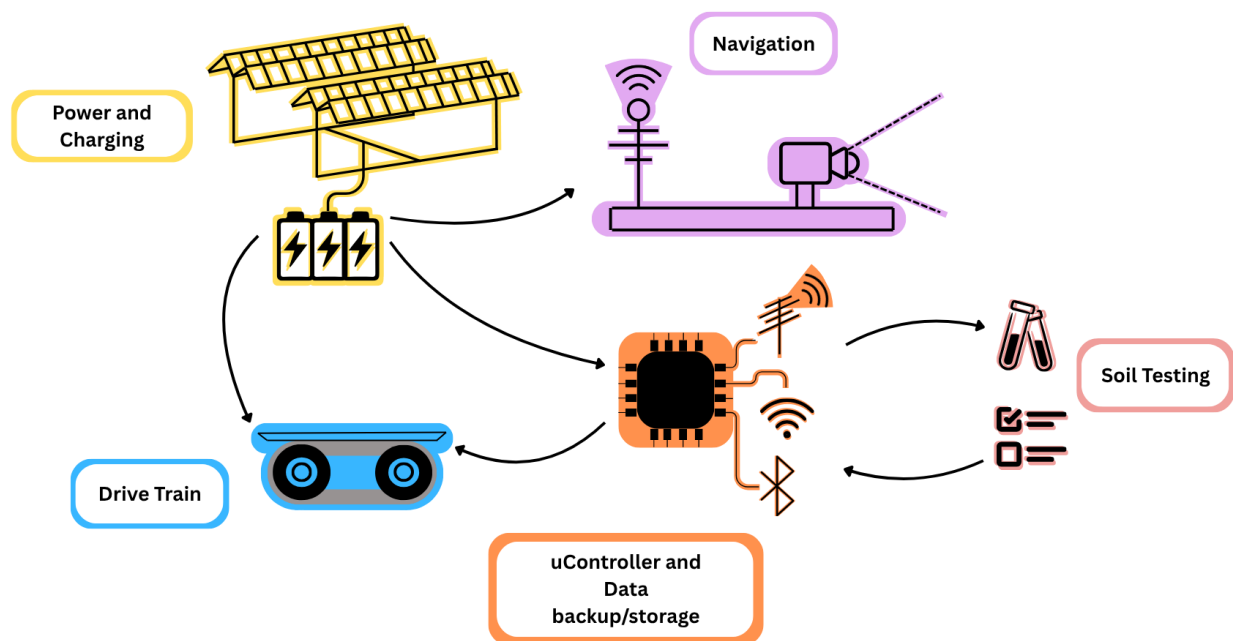
Approximately 25,000 people, including more than 10,000 children, die every day due to starvation. Agricultural imports in the U.S. exceed its exports by \$41 billion each year. Agricultural arms struggle to keep up with the growing demand for domestically grown food. 49% of U.S. crop fields struggle with poor health and resource concerns like nutrient imbalances. Agricultural labs generally rely on farmers or field agents for collecting and testing soil samples. This work is often monotonous, requiring 10 to 20 individual samples from a single management zone to ensure accuracy. All this time spent collecting and transporting samples would be better spent inside the lab, analyzing the soil studies. Using farmers to collect samples is bound by human ability. Mistakes are easily made, soil is collected from the wrong place, or the wrong amount of soil is collected. Work can only be done during the day when the sun is up, and workers are awake, decreasing efficiency by 50%.

Collecting and analyzing these soil samples is a highly repetitive task, and ideal for automation. These types of robots do exist and are used in crop fields, but they tend to be very expensive, ranging around \$54,000. A cheaper and more accessible option is needed to increase the amount of food people can access.

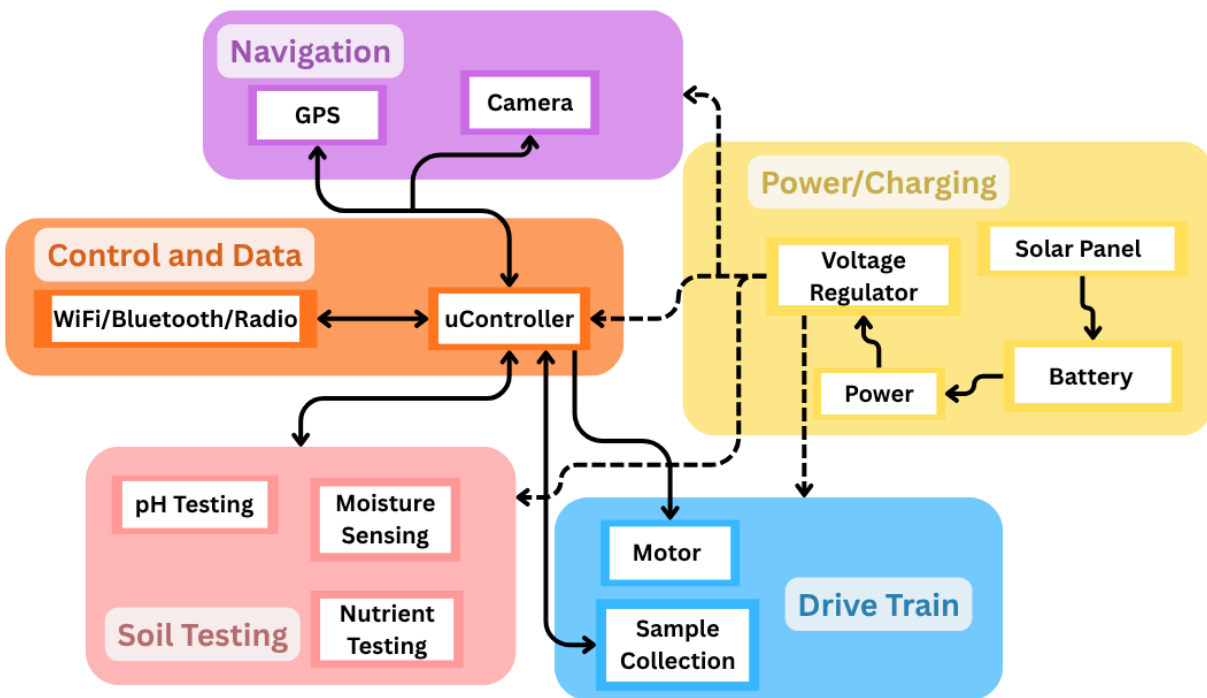
Proposed Solution:

A robot to help track crop field health by collecting and analyzing soil samples helps address this issue. Several subsystems help this robot operate independently, efficiently, and accurately. GPS navigation and drive training helps the robot travel in the correct zig zag pattern needed to ensure accuracy in the soil analysis. The robot takes each soil sample and looks for nutrient deficiencies or other signs of degradation, like an unbalanced pH level. This data is immediately backed up remotely as an extra safety feature in case something happens to the robot. The robot has a charging station that it returns to, as well as solar panels for an added source of power that helps it run longer.

Process Diagram:



System Diagram:





Team Charter

Dirty Dawgs

*D.I.R.T.T.
Lab*

<u>Department / Company</u>	Advisors	
CSE	Dr. Christopher Hudson	chudson@cse.msstate.edu
<i>Expected Meeting Time & Frequency w/ Advisor(s):</i> Tuesday, 3:30PM in Simrall Student Lounge		

Membership:

Project Manager	
<u>Nolan Brechtel</u>	
mdb209	CPE

Chief Scientist	
<u>George Utz</u>	
gdu11	EE

Other Team Members			
<u>Cara Belew</u>		<u>Zac Morrow</u>	
cgb331	E	zbm38	CPE
<u>Rebecca Ueltschey</u>			
rmu21	EE		

Mode & Frequency of Communication:

The team will mainly communicate via Microsoft Teams to communicate important information. We also have an iMessage group chat to exchange quick information. We plan to meet on Tuesdays at 3:30 PM in the Simrall Student Lounge, until we get a lab spot, then it will be moved there. If needed, we will begin meeting on Thursdays as well.



Submission Schedule:

Each team member will do their best to complete their section of the assignment that is due for the week before our team meeting each Tuesday, unless the workload requires more than one week. All documents due will be shared on teams with enough time before the due date that each person can submit it.

Team Roles:

Project Subsystems			
1	Navigation	Nolan Brechtel (lead)	Rebecca Ueltschey (support)
	<i>GPS and navigation</i>		
2	Soil Testing	Zac Morrow (lead)	Cara Belew (support)
	<i>Takes samples of soil and analyzes ph balance etc.</i>		
3	Power and Charging	George Utz (lead)	Zac Morrow (support)
	<i>Power system on robot (possibly solar) and charging station that robot returns to</i>		
4	Remote Data Backup	Cara Belew (lead)	George Utz (support)
	<i>Takes info gathered from soil testing and backs up data as it is gathered</i>		
5	Drive Train	Rebecca Ueltschey (lead)	Nolan Brechtel (support)
	<i>Building wheels/ mechanical aspect</i>		

Potential Obstacles:

This project may pose issues correctly analyzing the soil samples, and an external advisor who specializes in agriculture may be needed for us to know what data is most useful to collect from the soil.

One member is also a member of the MSU Debate Team, which requires travel 3-4 weekends per semester. Generally, this won't interfere with team meetings, but if travel occurs during a time the team needs to meet on a Thursday, the travelling member will Zoom in if necessary.

Conflict Resolution:

If there are any personal conflicts, such as team members disagreeing on workload, process of work, control, or communication, we have decided that the conflicting members will try to resolve the issue between themselves and then the other team members. If this cannot happen and the conflict begins interfering with the project, we will involve the faculty advisor, Dr. Christopher Hudson, to resolve the conflict. If this still does not mediate the conflict, we will involve Dr. Green and Mr. Robinson to resolve it.

If a member has a professional conflict, such as team members disagreeing on logistics, project direction, and part disagreement, they will bring it up to the team. We will discuss this conflict and see if we can find a solution that gets unanimous approval. If we cannot come to a unanimous decision, we will hold the decision to a vote, and an option must have 3 votes to win. If a vote happens and none of the options reach 3 votes after discussing and revoting, the project manager will decide.

If there are any technical conflicts, such as team members disagreeing on coding quality, data collection, and robot design, we will discuss this conflict and see if we can find a solution that gets unanimous approval. If we cannot come to a unanimous decision, we will hold the decision to a vote, and an option must have 3 votes to win. If a vote happens and none of the options reach 3 votes after discussing and revoting, the chief scientist will decide.

Additional Comments or Concerns:

[Signatures on the Next Page.]



Signatures

Team Signatures

Nolan Brechtel

Signature

Nolan Brechtel

Project Manager

9/8/2026

Date

George Utz

Signature

George Utz

Chief Scientist

9/8/2026

Date

Cara Belew

Signature

Cara Belew

Member

9/8/2026

Date

Zachary Morrow

Signature

Zachary Morrow

Member

9/8/2026

Date

Rebecca Ueltschey

Signature

Rebecca Ueltschey

Member

9/8/2026

Date

Advisor Signatures

Christopher R. Hudson

Signature

Christopher Hudson

CSE

Faculty Advisor

9/7/2026

Date